

# Off-Grid Communication with Meshtastic

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A pocket radio that texts your family for miles.  
No tower. No subscription. No license. No monthly bill.

## WHAT IT IS, IN ONE MINUTE

A small radio, a free app, and your phone. It sends **text and location**, not voice and not internet. It runs for days on a charge.

**Every radio repeats what it hears.** Your message hops from node to node, so you do not need to reach your neighbor directly. You just need to reach *somebody*. That is the mesh.

## WHAT TO BUY (PRICES SEPT 4, 2026)

| GET                               | WHY                                                                                                      | COST         |                                                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|
| <b>SenseCAP T1000-E</b>           | Meshtastic already installed. Charge it, pair it, done. Credit-card size, GPS, waterproof.               | <b>\$50</b>  |    |
| <b>Heltec V4 2-pack kit</b>       | <b>Best value.</b> Two complete nodes: batteries, cases, antennas. You and one other person, on the air. | <b>\$90</b>  |    |
| <b>Wio Tracker L1 Pro</b>         | Seeed's newer tracker. GPS, rechargeable. I carry two of these.                                          | <b>\$60</b>  |   |
| <b>SenseCAP Solar Node P1-Pro</b> | Solar + batteries, weatherproof. Put it on a pole and forget it. This is the one on Michael's pole.      | <b>\$140</b> |  |

**Two people on the mesh for about \$90.** That is less than a year of a satellite messenger subscription, and there is no bill after it.

## SAVE \$25 ON THE SOLAR NODE



Seeed direct

Buying the solar node straight from Seeed Studio runs about **\$115 delivered** instead of \$140. Worth it on the expensive one. On the cheap boards the shipping eats the savings, so Amazon is simpler there.

## THE ONE HARDWARE RULE

**Screw the antenna on BEFORE you power it up.**  
Transmitting with no antenna can damage the radio.  
Everything else is forgiving.

## SETUP, START TO FINISH

1. Charge it.
2. Update it: drag-and-drop (T1000-E) or `flasher.meshtastic.org` in Chrome (Heltec). **Do not tick "full erase."** It wipes your radio's identity keys, and your private messages to everyone who already knows you stop working until you both delete each other and start over. A normal update keeps them.
3. Install the free Meshtastic app (iOS or Android) and pair over Bluetooth.
4. **Check two settings: Region = US, and Preset = LongFast.** Wrong region and it will not transmit at all. Wrong preset and it transmits into an empty room. Both look exactly like a dead radio. These two are the first things to check, always.
5. Name your node something you will recognize. You are now on the mesh.



Full setup guide



Update your radio



Local mesh map



Local community



Free ham study

DO THIS TODAY

**Buy two and play. Leave one at home and see how far you get, or keep one in the car for emergencies.**

There are two separate questions, and mixing them up causes almost every problem.

1. Can our radios hear each other? That is **region + preset + frequency**.

2. Can we read what we hear? That is the **channel name and its key**.

Same room, different language: you hear nothing useful. Different room: nothing helps. **Get the radio settings right first.**

## RENAMING THE MAIN CHANNEL MOVES YOUR FREQUENCY

**RENAMING YOUR MAIN CHANNEL CHANGES YOUR FREQUENCY.** Meshtastic takes a hash of your **PRIMARY** channel's name and uses it to pick the actual frequency slot. Rename it and your radio quietly moves off the frequency your neighbors are on. Not a different chat room. A different frequency. No key will fix it.

So if you want privacy, **do not rename your main channel**. Read on.

## HOW TO GET A PRIVATE FAMILY CHANNEL (THE RIGHT WAY)

1. **Leave the main channel exactly as it came.** No name, default key, LongFast, region US. That is what keeps you on the mesh with everybody else.
2. **Add a second channel.** Give it a name and generate a **random** key. (Not "default" and not the "simple" ones. Those are published and everyone has them.)
3. **Share it by QR code** to the phones you trust. That is the whole handshake.

Now you have both: the public channel for reach, and a private channel only your family can read. You get 8 channels total, so there is room.

**Careful:** that QR is the key. Anyone who photographs it is in. Do not put it on a slide or a Facebook post.

## "IS LONGFAST PRIVATE?" NO.

LongFast is the channel every radio ships with, and its key is published in the documentation. Everyone on earth has it. Treat it like the public square: good for hello, for coordinating, for finding people. Wrong for anything you would not say out loud at the expo.

Keep it anyway. It is how you find your neighbors, and how they find you.

## DO MY MESSAGES STILL GET THROUGH IF WE ARE NOT ON THE SAME CHANNEL?

**Yes, and this is the best part.** A radio that cannot read your message still **carries** it. The address on the outside is readable; only the contents are locked. So strangers' nodes pass your private family messages along without ever being able to read a word of them.

Two things follow:

- Your private channel gets the range of *everybody's* radios.
- Owning a node is a contribution. Yours is carrying messages for people you have never met, right now, sitting in a window.

Leave the rebroadcast setting on **ALL** (the default) and your role on **CLIENT**. Do not set yours to Router or Repeater because it sounds stronger. Those are for nodes up high with real power, and a badly placed one makes the mesh worse.

## IS ANYONE OUT HERE?

**Yes.** Our regional network is **LZMesh** (Southwest Missouri), running around **60 nodes on a given day**. Map and Discord QR codes are on the front.

[map.lzmesh.com](http://map.lzmesh.com) • [discord.lzarc.com](https://discord.lzarc.com)

**Good news: they do it the way we just told you to.** LZMesh leaves the default channel alone and adds theirs as a second channel. So your brand-new radio is *already on their frequency* and already hears them. Add their channel from their site when you want to read that traffic too.

**TWO CAUTIONS.** Scanning their channel QR can **REPLACE CHANNELS YOU ALREADY HAVE**, so save a private one first. And if your node list looks empty, check **REGION** and **PRESET** before anything else. A brand-new board here came up on LongTurbo instead of LongFast: that put it on **908.750 MHZ** while everything else sat on **906.875**. It found nobody, on the same table as three working radios.

## WHEN YOU WANT MORE

- **MeshCore:** same radios, planned repeaters instead of everyone-repeats. Tighter at scale, more setup. [meshcore.co.uk](http://meshcore.co.uk)
- **Reticulum:** a whole encrypted network stack over any radio. The deep end. [reticulum.network](http://reticulum.network)
- **Ham radio:** voice, real power, repeaters. The Technician license is a weekend of study. Study **free at hamstudy.org**, built by our friend Richard Bateman (KD7BBC). The FCC charges **\$35** for the license itself. Studying costs nothing. Meshtastic is the on-ramp. Ham is the highway. Plenty of us run both.